

Datasets for the Mini-Project

Your Group may choose any ONE of the following datasets for the Mini-Project.

The exact Data Science problem that you define on the dataset may be different for every group, even if the dataset is the same. You may perform Regression or Classification, whichever you like, or a combination of techniques (even something new, not covered in class), on the dataset of your choice. The initial exploration of the data, visualization, and pre-processing will most likely differentiate your approach based on the problem. In some cases, you will also need data collection, curation, extraction of structured data, or joining of multiple datasets in the pre-processing phase. This will also depend on the problem of your choice.

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Dataset 1 : The Movie Database

Industrial Relevance – This datasets reflects how you can combine sources of information on the internet, and is quite a fun one to be creative!

Source : [https://\(https://www.themoviedb.org/\).https://www.kaggle.com/tmdb/tmdb-movie-metadata](https://(https://www.themoviedb.org/).https://www.kaggle.com/tmdb/tmdb-movie-metadata)
(<https://www.kaggle.com/tmdb/tmdb-movie-metadata>).

Documentation : <https://developers.themoviedb.org/3/getting-started/introduction>
(<https://developers.themoviedb.org/3/getting-started/introduction>).

Define your own problem on any dataset you extract.

Primarily offers metadata and statistics about movies and shows.

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Dataset 2 : Computer Parts (CPUs and GPUs)

Industrial Relevance – This dataset is typical of the electronics industry, where industry leaders often make strategic decisions based on parts coming together for computer architecture and prices.

Source1: <https://www.kaggle.com/datasets/michaelbryantds/cpu-and-gpu-product-data>
(<https://www.kaggle.com/datasets/michaelbryantds/cpu-and-gpu-product-data>).

Source2: <https://www.kaggle.com/datasets/alanjo/ddr2ddr3ddr4ddr5-ram-benchmarks>
(<https://www.kaggle.com/datasets/alanjo/ddr2ddr3ddr4ddr5-ram-benchmarks>).

Source3: <https://www.kaggle.com/datasets/alanjo/cpu-benchmarks> (<https://www.kaggle.com/datasets/alanjo/cpu-benchmarks>).

Define your own problem by combining computer hardware performance and price from different datasets provided above.

These datasets contain performance parameters, detailed specifications, release dates, and release prices of three main computer parts: Central Processing Units (CPU), Graphics Processing Units (GPU), and Random-access memory (RAM).

Dataset 3 : Worldwide Energy Usage

Industrial Relevance – Understanding and assimilating information about energy will define solutions towards sustainability, necessary for most businesses today.

Source : <https://github.com/owid/energy-data> (<https://github.com/owid/energy-data>)

Documentation : <https://ourworldindata.org/energy> (<https://ourworldindata.org/energy>)

Define your own problem on any dataset or subset you extract.

The Energy dataset provided (owid-energy-data.csv) is collected and maintained by Our World in Data, a regularly updated data that is on energy consumption (primary energy, per capita, and growth rates), energy mix, electricity mix and other relevant metrics. Metadata include time serial (year), country, population, GDP, generation, share, consumption of a plenty of energy sources (e.g., biofuel, coal, gas, fossil, hydrogen, carbon, oil, solar, wind, etc.), refer to owid-energy-codebook.csv for metadata description.

Dataset 4 : Global Supply Chain Logistics (Mining Industry)

Industry Relevance – Logistics is often time-sensitive and data analytics is a key part of the industry where engineers can contribute.

Source:

https://figshare.com/articles/dataset/Mining_Company_s_Global_Supply_Chain_Logistics_Data_for_a_Medium_Size_Excavator/1595938/4
(https://figshare.com/articles/dataset/Mining_Company_s_Global_Supply_Chain_Logistics_Data_for_a_Medium_Size_Excavator/1595938/4)

Documentation:



Original dataset desription.pdf

Define your own problem on any dataset you extract.

The dataset provided has been generated starting from a supply chain scenario of a company for a medium size excavator. Primarily offers data and metadata about production, transportation, costs, and tariffs of an excavator and its ten main components.

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Dataset 5 : Metamaterial Antennas

Industry Relevance – interesting technology applied in many relevant fields, from high-performance aircraft components to mobile communication and wireless connection. Complex

Source: <https://www.kaggle.com/datasets/renanmav/metamaterial-antennas>
(<https://www.kaggle.com/datasets/renanmav/metamaterial-antennas>).

Documentation: [Metamaterial Antennas info.docx \(https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40171711_1/xid-40171711_1\)](https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40171711_1/xid-40171711_1).

Define your own problem using data you find in source.

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Dataset 6 : Spam Dataset

Industry Relevance – cybersecurity, common application for data scientists

Source: [spambase.csv \(https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40172370_1/xid-40172370_1\)](https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40172370_1/xid-40172370_1).

Documentation: [spambase.names.txt \(https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40172371_1/xid-40172371_1\)](https://ntulearn.ntu.edu.sg/bbcswebdav/pid-5781307-dt-content-rid-40172371_1/xid-40172371_1).

Define your own problem using data you find in source.

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